Why latency alone isn't enough

- Latency tells us how long one kernel launch takes, but it depends on the problem size—and therefore on how much work was requested.
- To compare performance across problem sizes, we need to account for both the work completed and the time required. This gives us **achieved throughput**.
- Since hardware is characterized by its **peak throughput**, the two values provide a first gauge of how much compute capability our kernel realizes.

Throughput as a performance measure

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Throughput is the total useful work (FLOPs) done per unit time. *We previously discussed theoretical peak throughput of hardware.*

$$\text{Throughput} = \frac{\text{FLOPs}}{\text{time}}$$

Example (Car performance)

The trip took us 2 hours.

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Cal Poly Pomona $\rightarrow$ Mt. SAC

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Cal Poly Pomona $\rightarrow$ Mt. SAC

$$\frac{\sim 7 \text{ miles}}{2 \text{ hours}} \approx 3.5 \text{ mph}$$

Cal Poly Pomona $\rightarrow$ Cal Poly SLO

$$\frac{\sim 200 \text{ miles}}{2 \text{ hours}} \approx 100 \text{ mph}$$

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Example

Peak throughput of our RTX 3090 is 35.6 TFLOP/s. We observed 360ms or 0.36s of latency for $n = 8192$, where the total FLOPs required is 1.10 TFLOPs. Thus, we have an observed throughput of

$$\frac{1.10\text{TFLOPs}}{0.36\text{s}} \approx 3.05\text{TFLOP/s}.$$

Roughly 10% of peak throughput.

Arithmetic Intensity

Arithmetic Intensity

We define the **Arithmetic Intensity** (AI) of our kernel as the number of FLOPs done per bytes moved from DRAM.

$$\text{AI} = \frac{\text{FLOPs}}{\text{bytes moved}}$$

Example

For $n, k, m = 8192$, our tiled kernel moved a total of 137.7 GB of data from DRAM. With a total work count of 1.10 TFLOP, its arithmetic intensity is

$$\text{AI} = \frac{1.10 \times 10^{12} \text{ FLOP}}{137.7 \times 10^{9} \text{ bytes}} \approx \boxed{8.0 \text{ FLOP/byte}}.$$

Roofline Model

Roofline Model

For a given system, we can predict a the **upper bound** of throughput of a kernel given its AI. Up to the **ridge point**, we are **memory bound**, increasing AI will raise the upper bound of throughput. However, at the **ridge point**, we become **compute bound**, and no longer benefit from increasing AI.

Example

For our 3090, we have 35.6 TFLOP/s of peak FP32 throughput and 936 GB/s of memory bandwidth.

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At AI $= 8.0$ FLOP/byte, we are below the ridge point, and are still memory bound.

Roofline Model of our RTX 3090

